Shuchir Harsha

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EDUCATION

University of California at San Diego — La Jolla, CA

Bachelor of Science in Computer Science and Mathematics — *Expected Graduation: June 2027*

- Courses: Algorithms, Advanced Data Structures, Discrete Mathematics & Graph Theory, Computer Organization & Assembly, Digital Design, Probability & Statistics, Linear Algebra, Numerical Analysis
- Activities: Eagle Scout (led 60-person troop planning), Triton Robotics Embedded Team, Software & Computer Engineering Club

EXPERIENCE

Hewlett Packard Enterprise (Juniper Mist AI) — Sunnyvale, CA

Cloud Engineering Intern, Cloud Data Platform — *Jun 2026 - Present*

- Built application QoS statistics processing into a production telemetry pipeline on **Kafka, Apache Storm, and Elasticsearch**, computing **ACK RTT, TCP retransmissions, congestion window, throughput, and flow counts** per client and per application.
- Wrote an Apache **Storm bolt** that joins per-application traffic statistics with wireless client and radio telemetry into a single correlated record, so a slow application can be traced back to the wireless link, the wired network, or the WAN.
- Developed a **Java 11 Flink sink** that bulk indexes into Elasticsearch with checkpointing, retry handling, and **idempotent writes**, replacing a heavier client-based path. Covered it with integration tests against a mock HTTP server and deployed it to **AWS staging**.
- Added **root-cause scoring** to the QoS detector, labeling each degraded app with the metric driving the problem.
- Exposed **REST endpoints** aggregating Elasticsearch QoS data into per-app performance metrics for App Insights dashboards.
- Split the **Kafka-to-ORC archival path** into per-topic writers and renamed topics across AWS/GCP staging, coordinating **4 services**.
- Wrote **Python and Bash tooling** cross-checking Kafka, Elasticsearch, and S3 records against raw packet captures before releases.

NETGEAR — San Jose, CA

Software Engineer Intern — *Jun 2025 - Sep 2025*

- Built **Nimbis**, a Python LLM agent with **25+ MCP tools** for conversational control of cloud and on-prem NETGEAR devices.
- Paired LLM reasoning with **structured API validation** so network operations execute safely and deterministically.
- Wrote concurrent REST and JSON-RPC clients with **asyncio and httpx**, replacing sequential polling with parallel fleet queries.
- Designed a secure device layer with **dynamic API discovery**, session auth, and TLS, isolating credentials from tooling.
- Deployed **RAG on locally hosted LLMs**, keeping diagnostics on-premises so customer network data never left the environment.

NeuroLeap — San Jose, CA

Embedded Software Intern — *Apr 2024 - Oct 2024*

- Ported **Debian 11** onto **Android 12** with **proot and VNC** so the team could build gateway software with standard Linux tooling.
- Developed **UART and I2C** firmware code for microcontroller flashing and real-time peripheral control (7-segment displays).
- Set up embedded Linux build-and-test workflows for **Arduino and Raspberry Pi**, shortening firmware iteration cycles.

PROJECTS

netpilot | *Python, asyncio, httpx, Pydantic, MCP*

- Built an open-source **LLM agent** that diagnoses and configures network devices from plain English, exposed as an MCP server.
- Designed a **deterministic validation layer** with typed schemas and risk tiers so no invalid command reaches a device.

Object Detecting Camera | *Python, TensorFlow Lite, Embedded Linux*

- Deployed real-time object detection with **TensorFlow Lite** on embedded Linux, reaching **~99% accuracy** on common objects.

Autonomous Robot | *C++, Arduino, Ultrasonic Sensors*

- Implemented **C++ control logic** for ultrasonic sensor-based navigation, completing timed maze traversal challenges.

TECHNICAL SKILLS

Languages: Java, Python, Go, C++

Frameworks & Tools: Kafka, Storm, Flink, Elasticsearch, AWS, GCP, Embedded Linux, REST, JSON-RPC, MCP, pytest, Git